

Data-Driven and Simplified Crank–Nicolson Model for Heat Distribution in Habitat 1.0

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Research supported by Analog Astronaut Training Centre (AATC)[1] using sensor data from
Habitat 1.0 – EMMPOL25 Mission

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Abstract

Efficient thermal control is essential for comfort, safety and energy use inside closed habitats. During the EMMPOL25 analog mission at Habitat 1.0, temperature data was recorded in two rooms using Netatmo sensors[2] placed at different heights. In this work we develop a simple, data-driven model of heat distribution based on the heat diffusion equation. Physical parameters such as thermal diffusivity, heat transfer coefficient and heater power were estimated from warm-up and cool-down segments of the sensor data. The heat equation was then solved numerically in two dimensions using the Crank–Nicolson method. Simulation results reproduce the main thermal behaviour observed in the measurements and provide realistic spatial heat-distribution patterns. Validation against sensor data shows good agreement for the bedroom and larger discrepancies for the geolab, highlighting the limitations of the diffusion-only approach. This work provides a practical basis for future improvements toward more advanced habitat thermal control and energy planning.

Index Terms: Heat distribution, Thermal modelling, Crank–Nicolson, Data-driven simulation, Habitat environment, EMMPOL25, Thermal diffusivity, Heat transfer coefficient, Thermal control, Heat diffusion equation, Numerical heat transfer, Indoor temperature dynamics, Energy-efficient habitat design

1. Introduction

Thermal control is an important aspect of maintaining comfort, safety and energy efficiency in closed habitats. Small research stations, analog astronaut habitats and future extraterrestrial bases must operate with limited energy resources while ensuring stable environmental conditions for their occupants. Understanding how heat is distributed within habitat rooms can therefore support both operational planning and habitat design.

Analog missions provide a valuable opportunity to investigate environmental processes under controlled conditions without the cost and risk associated with real space missions. During the EMMPOL25 analog astronaut mission, conducted between 29 November and 6 December 2025 at Habitat 1.0 operated by the Analog Astronaut Training Centre (AATC), temperature measurements were continuously recorded using Netatmo sensors located in several rooms of the habitat. Multiple sensors were placed at different heights, allowing both temporal temperature evolution and potential vertical stratification effects to be investigated.

The objective of this work is to develop and validate a simple data-driven model of heat distribution inside Habitat 1.0. Rather than relying solely on literature values, the model parameters were estimated directly from measured temperature

data. Heating and cooling episodes were identified automatically from the sensor records, and characteristic thermal time constants were extracted by fitting exponential functions to the observed temperature evolution. These time constants were subsequently used to estimate effective thermal diffusivities and wall heat transfer coefficients for two rooms: the bedroom and the geolab.

The bedroom and geolab were selected because they provided complete sensor coverage and clearly identifiable heating cycles. The bedroom has dimensions of 5.05×5.05 m, while the geolab measures 2.25×6.35 m. The relatively small temperature differences observed between sensors located at 1 m and 2 m suggest that vertical temperature gradients were moderate during the analyzed periods. Consequently, a two-dimensional model was adopted as a first-order approximation.

Heat propagation inside each room was described using the two-dimensional heat diffusion equation with a localized heater source term. The governing equation was solved numerically using the Crank–Nicolson finite-difference method combined with Gauss–Seidel iteration. The resulting simulations produced spatial temperature distributions that could be visualized as heat maps and compared with measured temperature evolution.

It is important to emphasize that the present model intentionally represents a simplified description of reality. In indoor environments, heat transfer is strongly influenced by air circulation and convective transport. Since explicit airflow modelling was not included, the fitted thermal diffusivity should be interpreted as an *effective diffusivity* that captures both molecular diffusion and unresolved convective mixing processes. Consequently, the model is not intended to provide a complete physical description of room-scale thermal dynamics, but rather a computationally efficient approximation calibrated directly from experimental observations.

To evaluate the model, simulated average room temperatures were compared with measured sensor data. The comparison showed reasonable agreement for the bedroom, with a root-mean-square error (RMSE) of 1.16°C and a mean absolute error (MAE) of 1.11°C . Larger discrepancies were observed in the geolab (RMSE = 8.18°C , MAE = 7.78°C), indicating that additional physical processes, particularly convective heat transport and airflow patterns, play a more significant role in that room. These validation results provide valuable insight into both the capabilities and limitations of simplified diffusion-based thermal models.

The remainder of this paper presents the experimental data acquisition procedure, parameter estimation methodology, numerical model formulation, simulation results and validation against measurements. Finally, the limitations of the current

approach and possible directions for future model development are discussed.

2. Mission Context and Data Acquisition

This study was conducted during the EMMPOL25 analog astronaut mission, which took place between 29 November and 6 December 2025 at Habitat 1.0 operated by the Analog Astronaut Training Centre (AATC) in Rzepiennik Strzyżewski, Poland. Habitat 1.0 is a terrestrial analog facility designed to simulate selected operational conditions encountered in isolated and confined environments. Such habitats provide a controlled setting for testing technologies, procedures and scientific experiments relevant to future space missions.

The thermal environment inside the habitat was monitored using a network of Netatmo indoor air quality sensors. These devices recorded temperature measurements continuously throughout the mission and stored the results in CSV format. The collected data provided an opportunity to investigate the thermal behaviour of selected habitat rooms during normal crew operations and heating cycles.

Two rooms were selected for detailed analysis: the bedroom and the geolab. These rooms were chosen because they contained a sufficient number of sensors and exhibited clearly identifiable heating and cooling periods. The bedroom has dimensions of 5.05×5.05 m, while the geolab measures 2.25×6.35 m. In both rooms, sensors were installed at approximately 1 m and 2 m above the floor in order to investigate potential vertical temperature gradients caused by heating and air circulation.

For the bedroom, temperature measurements were obtained from six sensors. Three sensors were located at a height of approximately 1 m and three additional sensors were located at approximately 2 m. Similarly, eight sensors were used in the geolab, with four sensors positioned at each height level. The use of multiple sensors reduced the influence of individual sensor noise and allowed representative average temperatures to be calculated for each room.

The raw measurements were exported from the Netatmo platform as comma-separated value (CSV) files. Each file contained timestamps and temperature readings recorded at irregular intervals. Prior to analysis, all datasets were processed using Python. Temperature values were converted to numerical format, timestamps were transformed into a common datetime representation, and the measurements were resampled to a uniform 10-minute temporal resolution. This resampling step ensured that data originating from different sensors could be compared directly and combined into common room-level temperature series.

For each room, the average temperature at 1 m and 2 m was calculated as

$$T_{\text{avg}}(t) = \frac{1}{N} \sum_{i=1}^N T_i(t), \quad (1)$$

where N denotes the number of sensors at a given height and $T_i(t)$ is the temperature recorded by the i -th sensor. The resulting averaged time series provide a more robust representation of the thermal state of the room than individual sensor measurements.

To assess possible thermal stratification, the temperature difference between the upper and lower sensor levels was also examined:

$$\Delta T(t) = T_{2\text{m}}(t) - T_{1\text{m}}(t). \quad (2)$$

The observed differences were generally small throughout the analysed periods, indicating relatively effective air mixing inside both rooms. This observation supported the use of a two-dimensional thermal model as a first approximation of the heat transport process.

Figures 1 and ?? present the processed temperature measurements and the heating and cooling episodes identified from the data. These measurements form the basis for the parameter estimation procedure described in the following sections and provide the experimental reference against which the numerical simulations are evaluated.

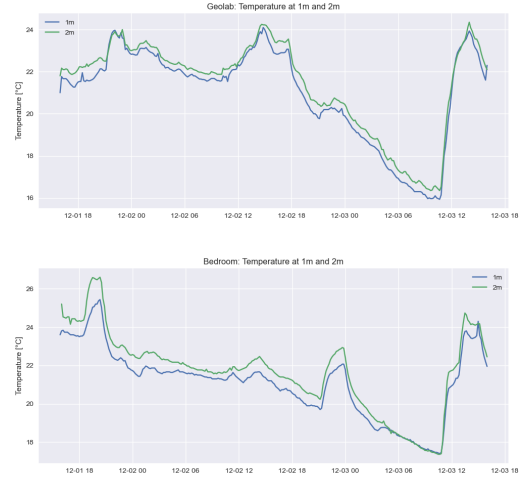


Figure 1: Average temperatures recorded in the geolab and bedroom at sensor heights of 1 m and 2 m. The data were resampled to a uniform 10-minute interval before analysis.

3. Heat Equation Model and Numerical Method

To investigate heat propagation inside Habitat 1.0, a simplified two-dimensional thermal model was developed. The objective of the model is not to reproduce every physical process occurring inside the rooms, but rather to capture the dominant thermal behaviour observed in the sensor measurements while maintaining low computational cost and a limited number of parameters.

The temperature field inside a room is represented by a continuous function $T(x, y, t)$ describing the air temperature at position (x, y) and time t . Heat transport is modeled using the two-dimensional heat diffusion equation with a localized heat source:

$$\frac{\partial T}{\partial t} = D \nabla^2 T + S(x, y), \quad (3)$$

where D denotes the thermal diffusivity, ∇^2 is the Laplace operator, and $S(x, y)$ represents the heat source associated with the room heater.

The heat source is assumed to occupy a small circular region corresponding to the physical heater location. Within this region the source term is constant and equal to S_{max} , while outside the heater region the source term is zero:

$$S(x, y) = \begin{cases} S_{\max}, & (x, y) \in \Omega_h, \\ \text{otherwise}, & 0, \end{cases} \quad (4)$$

where Ω_h denotes the heater area.

The present model intentionally neglects explicit convective transport. In indoor environments, heat transfer is strongly influenced by air circulation generated by buoyancy effects, ventilation and human activity. Since such processes are not modeled directly, the fitted diffusivity parameter should be interpreted as an effective diffusivity that accounts for both molecular diffusion and unresolved convective mixing.

3.1. Room Geometry and Computational Grid

The bedroom was modeled as a square domain with dimensions

$$L_x = L_y = 5.05 \text{ m}, \quad (5)$$

while the geolab was represented by a rectangular domain of dimensions

$$L_x = 2.25 \text{ m}, \quad L_y = 6.35 \text{ m}. \quad (6)$$

Each room was discretized using a uniform Cartesian grid with spatial step

$$\Delta x = \Delta y = 0.10 \text{ m}. \quad (7)$$

The resulting computational mesh stores one temperature value at each grid node. The simulation begins from a uniform initial temperature corresponding to the measured indoor temperature at the beginning of the analyzed heating episode.

Time integration is performed using a constant time step

$$\Delta t = 10 \text{ s}. \quad (8)$$

This temporal resolution was found to be sufficient for the relatively slow thermal processes observed inside the habitat.

3.2. Crank–Nicolson Discretization

Equation (3) was solved numerically using the Crank–Nicolson finite-difference method. This scheme is widely used for diffusion problems because it combines second-order accuracy in time with unconditional stability.

After discretization, the governing equation can be written as

$$T_{i,j}^{n+1} - \alpha \nabla^2 T_{i,j}^{n+1} = T_{i,j}^n + \alpha \nabla^2 T_{i,j}^n + \frac{\Delta t}{2} S_{i,j}, \quad (9)$$

where

$$\alpha = \frac{D, \Delta t}{2, \Delta x^2}. \quad (10)$$

Because temperatures at the new time level $n + 1$ appear on both sides of the discretized equation, the method is implicit and requires the solution of a linear system at every time step.

To obtain the temperature field, Gauss–Seidel relaxation was employed. Iterations were repeated until the residual dropped below a prescribed convergence tolerance of

$$10^{-8}. \quad (11)$$

The implicit formulation allows relatively large time steps to be used while maintaining numerical stability throughout the simulation.

3.3. Boundary Conditions

Heat exchange with the surroundings was modeled using a combination of fixed-temperature and convective boundary conditions.

Internal walls separating the analyzed room from the rest of the habitat were assumed to remain at the indoor reference temperature. These boundaries were implemented as Dirichlet conditions:

$$T = T_{\text{room}}. \quad (12)$$

External walls were modeled using Newton's law of cooling,

$$-D \frac{\partial T}{\partial n} = h (T - T_{\infty}), \quad (13)$$

where h is the heat transfer coefficient, T_{∞} denotes the outside temperature, and $\partial T / \partial n$ is the temperature gradient normal to the wall.

For the bedroom, the west and north walls were treated as internal walls, while the east and south walls were modeled as external boundaries subject to convective heat loss.

For the geolab, the west and east walls were treated as internal walls, whereas the north and south walls were modeled as external walls with convective heat exchange.

Outdoor temperatures during the mission varied approximately between -1°C and 6°C . Since the heating and cooling intervals used for parameter extraction lasted only a few hours, the outside temperature was approximated as constant within each analyzed episode. Consequently, the assumption of constant T_{∞} within a single analyzed episode was considered acceptable.

3.4. Model Parameters

The model requires three effective physical parameters:

- thermal diffusivity D ,
- wall heat transfer coefficient h ,
- heater strength $S_{\max}$.

These parameters were not taken from literature values. Instead, they were estimated directly from measured temperature data using the procedure described in the following section.

For the final simulations, the estimated parameter values were

$$D_{\text{bedroom}} = 1.82 \times 10^{-3} \text{ m}^2/\text{s}, \quad (14)$$

$$h_{\text{bedroom}} = 0.106 \text{ W}/(\text{m}^2\text{K}), \quad (15)$$

for the bedroom, and

$$D_{\text{geolab}} = 1.30 \times 10^{-3} \text{ m}^2/\text{s}, \quad (16)$$

$$h_{\text{geolab}} = 0.078 \text{ W}/(\text{m}^2\text{K}), \quad (17)$$

for the geolab.

The heater source strength was estimated as

$$S_{\max} = 5.77 \text{ K/s} \quad (18)$$

for both rooms.

The resulting model provides a computationally efficient framework for studying thermal behaviour inside Habitat 1.0 and forms the basis for the parameter estimation and validation procedures presented in the following sections.

4. Simulation Results and Heat Distribution

After estimating the effective thermal parameters from the sensor measurements, the Crank–Nicolson solver was used to simulate heat propagation in the bedroom and geolab. The simulations were initialized using the measured room temperature at the beginning of the selected heating episode and were subsequently evolved in time according to the heat diffusion equation described in Section 3.

To visualize the evolution of the temperature field, snapshots of the simulated temperature distribution were saved at selected times after heater activation and rendered as two-dimensional heat maps. All figures use the same temperature scale to facilitate direct comparison between rooms and between different simulation times.

4.1. General Characteristics of Heat Propagation

In both rooms, the heater acts as a localized source of thermal energy. Consequently, the earliest stages of heating are characterized by the formation of a warm region in the immediate vicinity of the heater. As time progresses, heat spreads throughout the room due to the diffusion process represented by Equation (3).

The simulation predicts three characteristic phases of temperature evolution:

1. **Local heating phase** — immediately after heater activation, temperature increases rapidly near the heat source while the rest of the room remains close to the initial temperature.
2. **Expansion phase** — the heated region grows and thermal gradients develop between the heater and more distant parts of the room.
3. **Approach to equilibrium** — temperature gradients gradually decrease as heat becomes more uniformly distributed throughout the room.

The resulting temperature fields provide insight into spatial heating patterns that cannot be obtained directly from a limited number of temperature sensors.

4.2. Bedroom Simulation

The bedroom was modeled as a square room with dimensions of 5.05×5.05 m. The west and north walls were treated as internal boundaries maintained at the indoor reference temperature, while the east and south walls were modeled as external boundaries subject to convective heat loss.

Figure 2 presents the simulated temperature distribution shortly after heater activation. During the first minute of heating, a compact region of elevated temperature develops around the heater location. The temperature gradient is steepest in the immediate vicinity of the source, indicating rapid local heat input.

As the simulation progresses to approximately five minutes after heater activation, the heated region expands significantly. Heat begins to spread across a larger fraction of the room and the temperature field becomes noticeably smoother. The highest temperatures remain concentrated near the heater, but the surrounding air experiences a measurable increase in temperature.

The relatively large dimensions of the bedroom allow heat to diffuse over a broad area before reaching the external boundaries. As a result, the temperature gradients remain moderate and the room warms relatively uniformly. This behaviour is consistent with the effective diffusivity estimated from the mea-

surement data and with the observed heating dynamics in the sensor recordings.

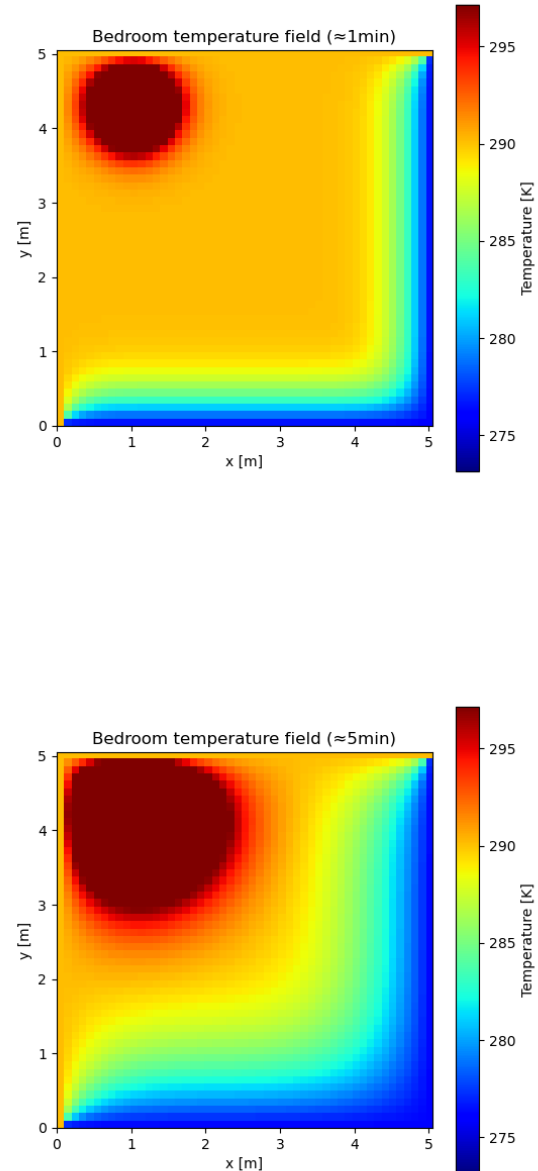


Figure 2: Simulated temperature distribution in the bedroom approximately 1 minute (top) and 5 minutes (bottom) after heater activation. The warm region expands gradually from the heater location and spreads throughout the room.

The simulation predicts that heat propagates relatively efficiently throughout the bedroom. The resulting temperature field is characterized by smooth gradients and a gradual transition between warmer and cooler regions. This behaviour suggests that the fitted effective diffusivity successfully captures the overall rate of thermal spreading observed experimentally.

4.3. Geolab Simulation

The geolab was modeled as a rectangular room with dimensions of 2.25×6.35 m. In contrast to the bedroom, the geolab contains two external walls located at the north and south boundaries, while the east and west walls were treated as internal boundaries.

The elongated geometry of the geolab has a noticeable influence on the simulated temperature distribution. Figure 3 shows the temperature field shortly after heating begins.

At early times, heating remains strongly localized near the source. The warm region initially occupies only a small portion of the room and pronounced temperature gradients are visible. Compared with the bedroom, the thermal field appears less uniform due to the elongated room geometry and the presence of two external walls through which heat is lost.

After approximately five minutes, heat has propagated further along the room length. However, substantial temperature differences remain between regions near the heater and those located farther away. The simulation therefore predicts a more heterogeneous temperature distribution than in the bedroom.

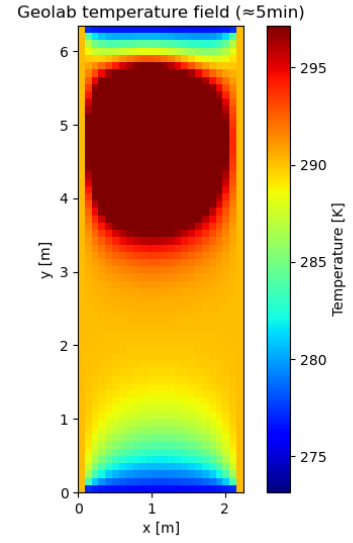
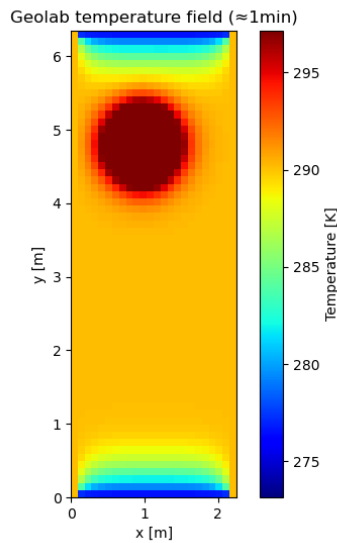


Figure 3: Simulated temperature distribution in the geolab approximately 1 minute (top) and 5 minutes (bottom) after heater activation. Heat spreads from the source but stronger temperature gradients persist because of the room geometry and external wall configuration.

The simulations indicate that heat propagation in the geolab is more sensitive to boundary conditions than in the bedroom. Because heat losses occur through both the north and south walls, the temperature field is influenced not only by the heater location but also by the interaction between heat generation and wall cooling. This effect becomes increasingly important as the simulation progresses.

4.4. Average Room Temperature Evolution

In addition to the spatial temperature distributions, the average temperature of each simulated room was recorded throughout the simulation. The average temperature provides a convenient scalar quantity that can be compared directly with measurements obtained from the sensor network.

Figure 4 shows the simulated average temperature evolution for both analyzed rooms. In both cases, the temperature increases rapidly after heater activation and subsequently approaches a quasi-steady state. The shape of the curves is consistent with the characteristic behaviour expected from a first-order thermal system and reflects the effective heating and cooling time constants extracted from the experimental data.

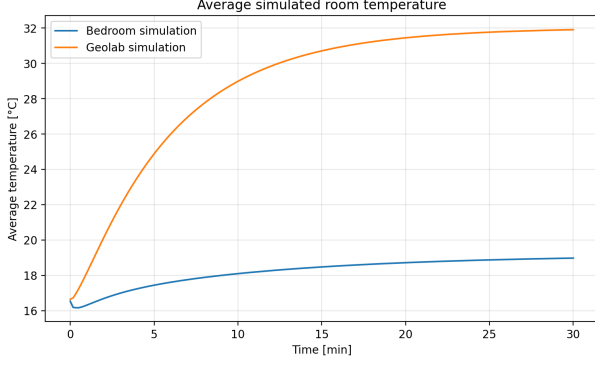


Figure 4: Average simulated temperature evolution in the bedroom and geolab during the analyzed heating episode.

The average-temperature curves provide a useful bridge between the spatially resolved numerical model and the room-scale measurements obtained from the sensor network. In the following section, these simulation results are compared directly with the measured temperatures in order to evaluate the performance of the model and identify its principal limitations.

5. Validation Against Measurements

A key objective of this study was to evaluate whether the simplified diffusion-based model is capable of reproducing the thermal behaviour observed inside Habitat 1.0. While visual inspection of temperature maps provides insight into the spatial evolution of heat, quantitative validation requires a direct comparison between simulated and measured temperatures.

To perform this comparison, the average temperature of each simulated room was computed at every time step and compared with the corresponding average temperature obtained from the sensor network. The measured temperature series were derived from the 1 m sensor level, which was used throughout the parameter estimation process. Simulated temperatures were interpolated onto the same temporal grid as the measurements to enable direct point-by-point comparison.

The agreement between simulation and measurements was quantified using the root-mean-square error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i^{\text{sim}} - T_i^{\text{obs}})^2}, \quad (19)$$

and the mean absolute error (MAE)

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |T_i^{\text{sim}} - T_i^{\text{obs}}|, \quad (20)$$

where T_i^{sim} and T_i^{obs} denote the simulated and measured temperatures, respectively, and N is the number of observations.

5.1. Bedroom Validation

Figure 5 compares the simulated and measured temperature evolution for the bedroom. Both curves exhibit a similar overall trend, characterized by a gradual temperature increase following heater activation and a subsequent approach toward a quasi-steady state.

The simulation slightly underestimates the measured temperature throughout most of the analyzed interval. Nevertheless, the heating dynamics, overall shape of the curve and characteristic time scale are reproduced reasonably well. The difference between the simulated and measured temperatures remains relatively small, indicating that the effective diffusivity and heat-loss parameters estimated from the data provide a realistic description of the room-scale thermal response.

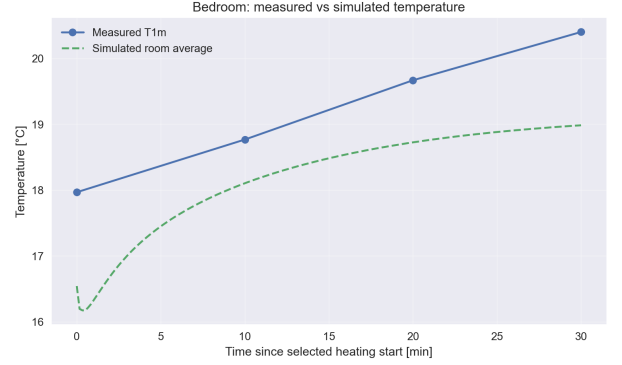


Figure 5: Comparison between measured and simulated temperature evolution in the bedroom during a representative heating episode.

For the bedroom, the quantitative error metrics are

$$\text{RMSE} * \text{bedroom} = 1.158^\circ\text{C}, \quad (21)$$

$$\text{MAE} * \text{bedroom} = 1.112^\circ\text{C}. \quad (22)$$

Considering the simplicity of the model and the absence of explicit airflow calculations, these values indicate satisfactory agreement between simulation and measurements.

5.2. Geolab Validation

Figure 6 presents the corresponding comparison for the geolab. While the simulation reproduces the general trend of increasing temperature after heater activation, substantially larger discrepancies are observed than in the bedroom.

The simulated temperature rises considerably faster and reaches significantly higher values than those recorded by the sensors. Although the model correctly predicts the qualitative behaviour of the system, it overestimates the heating rate and the final room temperature. The discrepancy increases progressively with time, indicating that important heat-transfer mechanisms are not fully captured by the diffusion-only formulation.

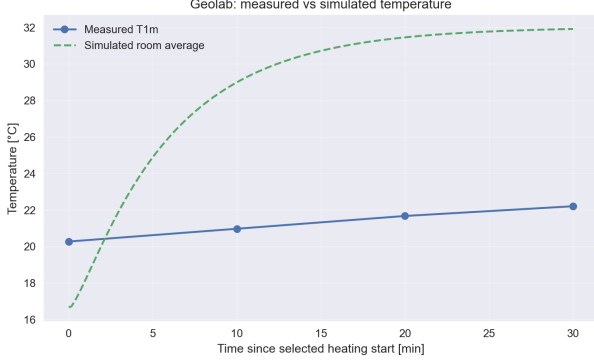


Figure 6: Comparison between measured and simulated temperature evolution in the geolab during a representative heating episode.

The corresponding error metrics are

$$\text{RMSE} * \text{geolab} = 8.177^{\circ}\text{C}, \quad (23)$$

$$\text{MAE} * \text{geolab} = 7.783^{\circ}\text{C}. \quad (24)$$

These values are substantially larger than those obtained for the bedroom and indicate that the current model is less successful in describing the thermal dynamics of the geolab.

5.3. Discussion of Model Accuracy

The validation results demonstrate that the predictive capability of the model depends strongly on the room being analyzed. For the bedroom, the agreement between measurements and simulations is reasonably good despite the simplified physical description. This suggests that the fitted effective parameters provide a useful approximation of the dominant thermal processes occurring within that room.

In contrast, the geolab exhibits significantly larger deviations. Several factors may contribute to this behaviour. First, the present model neglects explicit convective heat transport and assumes that all mixing processes can be represented through an effective diffusivity. Second, uncertainties in the heater source term and boundary conditions may have a stronger influence in the elongated geometry of the geolab. Third, occupant activity and air circulation patterns can produce temperature fluctuations that are not represented by the diffusion equation.

The results therefore support the interpretation of the model as an effective engineering approximation rather than a complete physical representation of indoor heat transfer. Nevertheless, the simulations successfully reproduce the principal heating behaviour and provide valuable insight into heat propagation patterns within Habitat 1.0.

Overall, the validation confirms that the proposed methodology can provide realistic thermal predictions for simple habitat environments while simultaneously highlighting the limitations that should be addressed in future work. In particular, the large discrepancy observed for the geolab suggests that explicit treatment of convective transport and airflow may be necessary to achieve higher predictive accuracy.

6. Conclusion

This work presented a data-driven thermal model of two rooms inside Habitat 1.0 based on temperature measurements collected during the EMMPOL25 analog astronaut mission. Using

data recorded by a network of Netatmo sensors, characteristic heating and cooling time constants were extracted and used to estimate effective thermal diffusivities, wall heat transfer coefficients and heater source strength. These parameters were subsequently incorporated into a two-dimensional heat diffusion model solved numerically using the Crank–Nicolson method.

The resulting simulations provided both temporal and spatial information about heat propagation within the habitat. Temperature maps revealed the formation and expansion of heated regions around the heater and demonstrated how room geometry and boundary conditions influence the distribution of thermal energy. The bedroom exhibited relatively uniform heat spreading, whereas the elongated geometry and increased external wall exposure of the geolab produced stronger temperature gradients and more pronounced heat-loss effects.

A direct comparison between simulated and measured temperatures was performed in order to evaluate the predictive capability of the model. For the bedroom, the simulation reproduced the observed heating behaviour with reasonable accuracy, yielding an RMSE of 1.158°C and an MAE of 1.112°C . This level of agreement demonstrates that a simple diffusion-based model calibrated from measurement data can provide useful estimates of room-scale thermal dynamics. In contrast, substantially larger errors were observed for the geolab, where the simulation overestimated the heating rate and reached higher temperatures than those recorded experimentally. The resulting RMSE of 8.177°C and MAE of 7.783°C indicate that additional physical processes influence heat transport in this room.

The observed differences between the two rooms highlight both the strengths and limitations of the proposed approach. The model successfully captures the dominant thermal behaviour and provides realistic qualitative predictions of heat distribution. At the same time, the validation results confirm that representing all air-mixing processes through an effective diffusivity is not always sufficient to reproduce measured temperatures accurately. In particular, natural and forced convection, airflow patterns, occupant activity and additional heat-loss mechanisms can significantly affect thermal dynamics and are not explicitly represented in the current formulation.

Despite these limitations, the presented methodology offers a practical and computationally efficient framework for analyzing thermal behaviour in small habitat environments. The approach requires only readily available temperature measurements and can be adapted to other rooms, missions or experimental facilities. As such, it provides a useful first step toward more advanced thermal simulations for analog astronaut habitats and future space habitation systems.

Future work should focus on extending the model beyond pure diffusion by incorporating convective heat transport, three-dimensional effects and more realistic airflow descriptions. Such developments would improve predictive accuracy and reduce reliance on empirically fitted effective parameters. Nevertheless, the results obtained in this study demonstrate that even a simplified data-driven model can provide valuable insight into heat propagation processes and serve as a foundation for future habitat thermal-control research.

7. References

- [1] Analog Astronaut Training Centre, “Analog habitat 1.0 and emmpol25 mission context,” Mission and facility information, 2025.
- [2] Netatmo, “Smart indoor air quality monitor,” Technical product documentation, 2024, accessed during the project preparation.